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Research Profile

My research develops the mathematics of hierarchical structure — non-Archimedean (p-adic) analysis, pseudodifferential operators, and statistical field theory — and, over the past five years, has applied it to the mathematical foundations of deep learning.

The deep learning program treats network architecture itself as a mathematical object rather than as a parameter count. Its principal results are: a rigorous correspondence between deep Boltzmann machines, deep belief networks, and p-adic statistical field theories; a universal approximation theorem for non-Archimedean deep networks, together with an isometric identification carrying the result back to real-valued functions on the interval; a dynamic mean-field theory for continuous random networks, with a criticality condition expressed through the largest Lyapunov exponent; and a rigorous thermodynamic limit for deep and recurrent networks in which the network admits a unique state inside a region of parameter space and infinitely many outside it — a genuine phase transition, whose critical organization in certain cases is described by a strange attractor.

That program rests on twenty-five years of work in analysis, number theory, and mathematical physics: Igusa local zeta functions, Newton polyhedra, and oscillatory integrals for meromorphic functions; Koba–Nielsen string amplitudes and their meromorphic continuation, including generalized Selberg–Mehta–Macdonald and Dotsenko–Fateev-type integrals; pseudodifferential equations over non-Archimedean spaces and the Markov processes they generate; and quantum mechanics over a discrete space, including the

identification of a broad class of p-adic Schrödinger equations as exact scaling limits of continuous-time quantum walks.

I have authored 3 research monographs (Cambridge University Press, Springer, De Gruyter) and more than 80 papers in journals spanning mathematics, mathematical physics, and computation (*Advances in Mathematics*, *Trans. Amer. Math. Soc.*, *J. London Math. Soc.*, *IMRN*, *J. Fourier Anal. Appl.*, *Reviews in Mathematical Physics*, *Nuclear Physics B*, *J. Phys. A*, *Annals of Physics*, *Physica D*, *J. High Energy Phys.*, *Nonlinearity*, *Commun. Number Theory Phys.*), and have supervised 17 Ph.D. dissertations in Colombia, Mexico, and the United States.

Citation metrics: ∼2,000 citations on Google Scholar (August 2026); see my Google Scholar profile.

Education

- D.Sc. Mathematics, 1996, National Institute of Pure and Applied Mathematics, IMPA, Rio de Janeiro, Brazil.
- M.Sc. Mathematics, 1992, Los Andes University, Bogotá D.C., Colombia.
- M.Sc. Electrical Engineering, 1992, Los Andes University, Bogotá D.C., Colombia.
- B.Sc. Mathematics, 1988, National University of Colombia, Bogotá D.C., Colombia.
- B.Sc. Electronics Engineering, 1988, Francisco José de Caldas University, Bogotá D.C., Colombia.

Appointments

- (12/2020 to present) Full Professor of Mathematics with tenure, the University of Texas Rio Grande Valley, United States.
- (2023-2025) Doctoral Faculty, Department of Physics, the University of Texas Rio Grande Valley, United States.

- (04/2016-12/2020) Research Professor level 3D at the Department of Mathematics, Center for Research and Advanced Studies of the National Polytechnic Institute, CINVESTAV, Mexico.
- (01/2013- 03/2016) Research Professor level 3C at the Department of Mathematics, Center for Research and Advanced Studies of the National Polytechnic Institute, CINVESTAV, Mexico.
- (05/2007-12/2013) Research Professor level 3B at the Department of Mathematics, Center for Research and Advanced Studies of the National Polytechnic Institute, CINVESTAV, Mexico.
- (03/2006- 04/2007) Associate Professor of Computer Science and Mathematics, Barry University, Department of Mathematics and Computer Science, Miami, U.S.A.
- (09/2001- 02/2006) Assistant Professor of Computer Science and Mathematics, Barry University, Department of Mathematics and Computer Science, Miami, Florida, U.S.A.
- (10/1997-09/2000) Director of Scientific Computation Laboratory, Autonomous University of Bucaramanga, Colombia.
- (08/1996-09/1997) Assistant Professor of Mathematics, Industrial University of Santander, Department of Mathematics, Bucaramanga, Colombia.
- (09/1991-09/1992). Research Engineer at Institute of Electronics and Communications, ITEC-TELECOM, Bogotá, Colombia.

Awards and Honors

- (2020–2029) Debnath Endowed Professorship in the Mathematical Sciences, the University of Texas Rio Grande Valley (reappointed 2023 and 2026).
- (2016–2025) Member of the National System of Researchers of Mexico (SNI), Investigador Nacional Nivel III (highest level). Nivel II (2012–2015); Nivel I (2007–2011).

- (2014) Regular Member of the Mexican Academy of Sciences.
- (2010) Alejandro Ángel Escobar Medal in Exact, Physical and Natural Sciences, Colombia.
- (2009) Regular Member of the Colombian Academy of Exact, Physical and Natural Sciences.

Grants

- Multiple grants for organizing meetings, conferences, and workshops in France, Germany, Spain, Mexico, and Colombia are not listed here. Please see the list of conferences that I have co-organized.
- (2020-2024) Group grant from Conacyt, Mexico, to conduct research in mathematical biology (Non-conventional mathematical and computational models for the study and analysis of relevant problems in biology). Multidisciplinary research grant.
- (2016-2020) Grant from Conacyt, Mexico, to conduct research in number theory/algebraic geometry, p-adic analysis and mathematical physics (Non-Archimedean Analysis and Geometry and Applications).
- (2011-2014) Grant from Conacyt, Mexico, to conduct research in number theory, algebraic geometry and p-adic analysis (Oscillatory Integrals and Exponential sums in Arithmetic and Complex Systems).
- (2005-2007) Young Investigator Grant, National Security Agency, US, to conduct research in number theory and p-adic analysis (Newton polyhedra, Oscillatory Integrals and Pseudo-Differential Operators Over p-adic Fields).
- Travel Grant from American Institute of Mathematics to attend the Workshop "Numerical Invariants of Singularities and Higher-dimensional Algebraic Geometry," Palo Alto, California, US, July 31st to August 4th, 2006.
- (2003-2004) Grant from Council of Grants and Research, Barry University, Miami, U.S.A, to conduct research in number theory and p-adic analysis (Local zeta functions and p-adic pseudo-differential equations).

- (2000-2002) Grant from COLCIENCIAS (Administrative Department of Science, Technology and Innovation), Colombia, to conduct research in number theory (Local Zeta Functions, non Degeneracy Conditions and Newton Polygons).
- (1998-2000) Grant from COLCIENCIAS (Administrative Department of Science, Technology and Innovation), Colombia, to conduct research in number theory (Local Zeta Functions of quasi-homogeneous polynomials).

Books

1. Zúñiga-Galindo W. A., "*p*-Adic Analysis: Stochastic Processes and Pseudo-Differential Equations." De Gruyter, 2025.
2. Khrennikov Andrei, Kozyrev Sergei, Zúñiga-Galindo W. A., "Ultrametric Pseudodifferential Equations and Applications". Encyclopedia of Mathematics and its Applications 168. Cambridge: Cambridge University Press (ISBN 978-1-107-18882-2), 2018.
3. Zúñiga-Galindo W. A., "Pseudodifferential Equations Over Non-Archimedean Spaces," Lectures Notes in Mathematics 2174, Springer, 2016. ISBN 978-3-319-46738-2.

Selected Publications

Grouped by theme; the complete chronological list follows.

Mathematical foundations of deep learning and neural networks

1. Zúñiga-Galindo, W. A. "Deep Neural Networks: A Formulation via Non-Archimedean Analysis," J. Fourier Anal. Appl. 32 (2026), no. 4, Paper No. 77. *(Universal approximation for hierarchical architectures.)*
2. Zúñiga-Galindo W. A., "Critical Organization of Deep Neural Networks, and *p*-Adic Statistical Field Theories," arXiv:2601.19070, under revision for J. Phys. A. *(Thermodynamic limit; phase transition.)*

3. Zúñiga-Galindo W. A., "Hierarchical connectivity sets the upper critical dimension of neural avalanches," arXiv:2610.36404, submitted to Physical Review Letters.

4. Zúñiga-Galindo W. A., "Dynamic mean-field theory for continuous random networks," J. Phys. A: Math. Theor. 58 (2025), 12520. *(Criticality via the largest Lyapunov exponent.)*

5. Zúñiga-Galindo, W. A. "A Correspondence Between Deep Boltzmann Machines and p-Adic Statistical Field Theories," Adv. Theor. Math. Phys. 28 (2024), no. 2, 679–741.

6. Zúñiga-Galindo W. A., He C., Zambrano-Luna B. A., "p-Adic statistical field theory and convolutional deep Boltzmann machines," Prog. Theor. Exp. Phys. 2023, no. 6, 063A01.

7. Zúñiga-Galindo, W. A., "p-Adic statistical field theory and deep belief networks," Physica A 612 (2023), Paper No. 128492.

8. Zúñiga-Galindo, W. A., "Non-Archimedean statistical field theory," Rev. Math. Phys. 34 (2022), no. 8, Paper No. 2250022.

9. Zúñiga-Galindo, W. A., and Zambrano-Luna B. A., "Hierarchical Wilson–Cowan Models and Connection Matrices," Entropy 25 (2023), no. 6, 949. *(Cortical dynamics compatible with small-world connectivity.)*

10. Zambrano-Luna B. A., Zúñiga-Galindo W. A., "p-Adic cellular neural networks: applications to image processing," Physica D 446 (2023), Paper No. 133668.

11. Zúñiga-Galindo W. A., Zambrano-Luna B. A., Indoung C., "Pattern Formation in Quantum Hierarchical Cellular Neural Networks," Commun. Nonlinear Sci. Numer. Simul. 163 (2026), Paper No. 110715.

Zeta functions, oscillatory integrals, and string amplitudes

11. Veys W., Zúñiga-Galindo W. A., "Zeta Functions and Oscillatory Integrals for Meromorphic Functions," Adv. Math. 311 (2017), 295–337.

12. Veys W., Zúñiga-Galindo W. A., "Koba–Nielsen local zeta functions, convex subsets, and generalized Selberg–Mehta–Macdonald and Dotsenko–Fateev-like integrals," Commun. Number Theory Phys. 20 (2026), no. 2, 353–390.

13. Zúñiga-Galindo, W. A., "Local Zeta Functions Supported on Analytic Sets and Newton Polyhedra," Int. Math. Res. Not. IMRN 2009, no. 15, 2855–2898.

14. Bocardo-Gaspar M., Veys W., Zúñiga-Galindo W. A., "Meromorphic continuation of Koba–Nielsen string amplitudes," J. High Energy Phys. 2020, 138.

15. Saia M. J., Zúñiga-Galindo W. A., "Local zeta functions for curves, non-degeneracy conditions and Newton polygons," Trans. Amer. Math. Soc. 357 (2005), 59–88.

Pseudodifferential equations, Markov processes, and complex systems

16. Khrennikov A., Kozyrev S., Zúñiga-Galindo W. A., "Ultrametric Pseudodifferential Equations and Applications," Encyclopedia of Mathematics and its Applications 168, Cambridge University Press, 2018.

17. Torba S. M., Zúñiga-Galindo W. A., "Parabolic Type Equations and Markov Stochastic Processes on Adeles," J. Fourier Anal. Appl. 19 (2013), no. 4, 792–835.

18. Zúñiga-Galindo W. A., "Non-Archimedean Reaction-Ultradiffusion Equations and Complex Hierarchic Systems," Nonlinearity 31 (2018), no. 6, 2590–2616.

19. Zúñiga-Galindo W. A., "Reaction-diffusion equations on complex networks and Turing patterns, via p-adic analysis," J. Math. Anal. Appl. 491 (2020), no. 1, 124239.

Quantum mechanics over a discrete space

20. Zúñiga-Galindo W. A., Mayes N. P., "p-Adic quantum mechanics, infinite potential wells, and continuous-time quantum walks," Rev. Math.

Phys. (2026), Paper No. 2550027.

21. Zúñiga-Galindo, W. A., "The p-Adic Schrödinger equation and the two-slit experiment in quantum mechanics," Ann. Physics 469 (2024), Paper No. 169747.

Papers Published

1. Zúñiga-Galindo W. A., Zambrano-Luna B. A., and Indoung, Chayapuntika. "Pattern Formation in Quantum Hierarchical Cellular Neural Networks."Communications in Nonlinear Science and Numerical Simulation 163 (2026), part 5, Paper No. 110715.

2. Zúñiga-Galindo W. A., Zambrano-Luna B. A., and Indoung, Chayapuntika. "Pseudo-Traveling Waves and Bumps in Quantum and Classical Hierarchical Cellular Neural Networks. "Journal of Pseudo-Differential Operators and Applications 17, 65 (2026).

3. Zúñiga-Galindo, W. A., Mayes, Nathaniel P." p-Adic quantum mechanics, infinite potential wells, and continuous-time quantum walks," Reviews in Mathematical Physics 38 (2026), no. 6, Paper No. 2550027, 32 pp.

4. Zúñiga-Galindo, W. A. "Deep Neural Networks: A Formulation via Non-Archimedean Analysis," J. Fourier Anal. Appl. 32 (2026), no. 4, Paper No. 77.

5. Veys Willem, Zúñiga-Galindo W. A., "Koba-Nielsen local zeta functions, convex subsets, and generalized Selberg-Mehta-Macdonald and Dotsenko-Fateev-like integrals," Commun. Number Theory Phys. 20 (2026), no. 2, 353–390.

6. Zúñiga-Galindo, W.A., "Quantum mechanics, non-locality, and the space discreteness hypothesis," Ann. Physics 489 (2026), Paper No. 170459.

7. Zúñiga-Galindo W. A., "2-Adic Quantum Mechanics, Continuous-Time Quantum Walks, and the Space Discreteness." Fortschr. Phys. 73 (2025), no. 8, Paper No. e70019.

8. Zúñiga-Galindo W. A., "Dynamic mean-field theory for continuous random networks," 2025 J. Phys. A: Math. Theor. 58 12520.

9. Zúñiga-Galindo W. A., "A Correspondence Between Deep Boltzmann Machines and p-Adic Statistical Field Theories," Adv. Theor. Math. Phys. Volume 28, Number 2, pp. 679-741 (2024).

10. Zúñiga-Galindo W.A., Zambrano-Luna B.A. & Dibba B., "Hierarchical Neural Networks, p-Adic PDEs, and Applications to Image Processing," J Nonlinear Math Phys 31, 63 (2024).

11. Zúñiga-Galindo W. A., "The p-Adic Schrödinger equation and the two-slit experiment in quantum mechanics," Ann. Physics 469 (2024), Paper No. 169747.

12. Zúñiga-Galindo W. A., "p-Adic Quantum Mechanics, the Dirac Equation, and the violation of Einstein causality," J. Phys. A: Math. Theor. 57 305301 (2024).

13. Chacón-Cortés L. F., Garcia-Bibiano C. A., Zúñiga-Galindo W. A.,"Turing Patterns in a p-Adic FitzHugh-Nagumo System on the Unit Ball," p-Adic Num Ultrametr Anal Appl 15, 245–265 (2023).

14. Zúñiga-Galindo, W. A., and B. A. Zambrano-Luna. 2023. "Hierarchical Wilson–Cowan Models and Connection Matrices," Entropy 25, no. 6: 949. https://doi.org/10.3390/e25060949.

15. Zúñiga-Galindo W. A., He. C., Zambrano-Luna B. A., "p-Adic statistical field theory and convolutional deep Boltzmann machines," Progress of Theoretical and Experimental Physics, Volume 2023, Issue 6, June 2023, 063A01, https://doi.org/10.1093/ptep/ptad061.

16. Zambrano-Luna B., Zúñiga-Galindo W. A., "p-Adic Cellular Neural Networks," J. Nonlinear Math. Phys. 30 (2023), no. 1, 3470. 11.

17. Zambrano-Luna, B. A., Zúñiga-Galindo, W. A., "p-adic cellular neural networks: Applications to image processing," Phys. D 446 (2023), Paper No. 133668.

18. Zúñiga-Galindo, W. A., "p-adic statistical field theory and deep belief networks," Phys. A 612 (2023), Paper No. 128492, 23 pp.

19. Chacón-Cortés L. F., Garcia-Bibiano C. A., Zúñiga-Galindo, W. A., "Local Well-Posedness of the Cauchy Problem for a p-Adic Nagumo-Type Equation," p-Adic Numbers Ultrametric Anal. Appl. 14 (2022), no. 4, 279–296.

20. Zúñiga-Galindo W. A., "Non-Archimedean quantum mechanics via quantum groups," Nuclear Phys. B 985 (2022), Paper No. 116021, 21 pp.

21. Zúñiga-Galindo, W. A.,"Non-Archimedean statistical field theory," Rev. Math. Phys. 34 (2022), no. 8, Paper No. 2250022, 41 pp.

22. Zúñiga-Galindo W. A., "Eigen's paradox and the quasispecies model in a non-Archimedean framework," Phys. A 602 (2022), Paper No. 127648, 18 pp.

23. Zúñiga-Galindo W. A. p-adic analysis: a quick introduction. p-adic analysis, arithmetic and singularities, 177–221, Contemp. Math., 778, Amer. Math. Soc., [Providence], RI, [2022], ©2022.

24. Zúñiga-Galindo, W. A., "Ultrametric diffusion, rugged energy landscapes and transition networks," Phys. A 597 (2022), Paper No. 127221, 19 pp.

25. Fuquen-Tibatá A. R., García-Compeán H., Zúñiga-Galindo W. A., Euclidean quantum field formulation of p-adic open string amplitudes, Nuclear Phys. B 975 (2022), Paper No. 115684, 27 pp.

26. Zúñiga-Galindo W. A., Zambrano-Luna B. A., León-Cardenal E., "Graphs, local zeta functions, log-Coulomb gases, and phase transitions at finite temperature," J. Math. Phys. 63 (2022), no. 1, Paper No. 013506, 21 pp.

27. Zúñiga-Galindo W.A., "Non-Archimedean Models of Morphogenesis." In: Zúñiga-Galindo W.A., Toni B. (eds) Advances in Non-Archimedean Analysis and Applications. STEAM-H: Science, Technology, Engineering, Agriculture, Mathematics & Health. Springer, Cham.(2021).

28. Bocardo-Gaspar M., García-Compeán H., López, E.Y., Zúñiga-Galindo W.A., "Local Zeta Functions and Koba–Nielsen String Amplitudes," Symmetry 2021, 13, 967.

29. Bocardo-Gaspar M., Veys W. & Zúñiga-Galindo W.A., "Meromorphic continuation of Koba-Nielsen string amplitudes," J. High Energ. Phys. 2020, 138 (2020).

30. Zúñiga-Galindo W. A., "Reaction-diffusion equations on complex networks and Turing patterns, via p-adic analysis," J. Math. Anal. Appl. 491 (2020), no. 1, 124239, 39 pp.

31. Zúñiga-Galindo W. A., Torba, Sergii M., "Non-Archimedean Coulomb gases," J. Math. Phys. 61 (2020), no. 1, 013504, 16 pp.

32. García-Compeán H., López Edgar Y., Zúñiga-Galindo W. A., "p-adic open string amplitudes with Chan-Paton factors coupled to a constant B-field," Nuclear Phys. B 951 (2020), 114904, 33 pp.

33. Arroyo-Ortiz Edilberto, Zúñiga-Galindo W. A., "Construction of p-adic Covariant Quantum Fields in the Framework of White Noise Analysis". Rep. Math. Phys. 84 (2019), no. 1, 1–34.

34. Bocardo-Gaspar M, Garcia-Compean H., Zúñiga-Galindo W. A., "Regularization of p-adic String Amplitudes, and Multivariate Local Zeta Functions," Lett. Math. Phys. 109 (2019), no. 5, 1167–1204.

35. Mendoza-Martínez M. L., Vallejo-Rodríguez J. A., Zúñiga-Galindo W. A., "Acausal quantum theory for non-Archimedean scalar fields," Rev. Math. Phys. 31 (2019), no. 4, 1950011, 46 pp.

36. Zúñiga-Galindo W. A., "Non-Archimedean Replicator Dynamics and Eigen's Paradox,"J. Phys. A 51 (2018), no. 50, 505601, 26 pp.

37. Torresblanca-Badillo Anselmo, Zúñiga-Galindo W. A.," Ultrametric diffusion, exponential landscapes, and the first passage time problem," Acta Appl. Math. 157 (2018), 93–116.

38. Bocardo-Gaspar M., Garcia-Compean H., Zúñiga-Galindo W. A., "p-Adic string amplitudes in the limit p approaches to one," J. High Energy Phys. 2018, no. 8, 043, front matter+22 pp.

39. Zúñiga-Galindo W. A., "Non-Archimedean Reaction-Ultradiffusion Equations and Complex Hierarchic Systems," Nonlinearity 31 (2018), no. 6, 2590-2616.

40. Bocardo-Gaspar Miriam, Zúñiga-Galindo W. A., "Local Zeta Functions for Rational Functions and Newton Polyhedra," in Singularities and Foliations. Geometry, Topology and Applications. BMMS 2/NBMS 3, Salvador, Brazil, 2015, Araujo dos Santos, R.N., Menegon Neto, A., Mond, D., Saia, M.J., Snoussi, J. (Eds.), p. 363-386, Springer Proceedings in Mathematics & Statistics.

41. Torresblanca-Badillo A., Zúñiga-Galindo W. A., "Non-Archimedean Pseudodifferential Operators And Feller Semigroups," p-Adic Numbers Ultrametric Anal. Appl. 10 (2018), no. 1, 57-73.

42. Chacón-Cortés L. F., Zúñiga-Galindo W. A., "Heat traces and spectral zeta functions for p-adic Laplacians," Algebra i Analiz, 29 (2017), no. 3, 144–166. Reprinted in St. Petersburg Math. J. 29 (2018), no. 3, 529–544.

43. Zúñiga-Galindo W. A., "Local zeta functions, pseudodifferential operators and Sobolev-type spaces over non-Archimedean local fields," p-Adic Numbers Ultrametric Anal. Appl. 9 (2017), no. 4, 314–335.

44. Veys Willem, Zúñiga-Galindo W. A., "Zeta Functions and Oscillatory Integrals for Meromorphic Functions," Adv. Math. 311(2017), 295–337.

45. Zúñiga-Galindo W. A., "Non-Archimedean white noise, pseudodifferential stochastic equations, and massive Euclidean fields," J. Fourier Anal. Appl. 23 (2017), no. 2, 288–323

46. Chacón-Cortes L. F., Zúñiga-Galindo W. A., "Non-local operators, non-Archimedean parabolic-type equations with variable coefficients and Markov processes," Publ. Res. Inst. Math. Sci. 51 (2015), no. 2, 289–317.

47. Zúñiga-Galindo W. A., "The non-Archimedean stochastic heat equation driven by Gaussian noise," J. Fourier Anal. Appl. 21 (2015), no. 3, 600–627.

48. Zúñiga-Galindo, W. A., "The Cauchy problem for non-Archimedean pseudodifferential equations of Klein–Gordon type," J. Math. Anal. Appl. 420 (2014), no. 2, 1033–1050.

49. Casas-Sánchez O. F., Zúñiga-Galindo, W. A., "p-adic elliptic quadratic forms, parabolic-type pseudodifferential equations with variable coefficients and Markov processes," p-Adic Numbers Ultrametric Anal. Appl. 6 (2014), no. 1, 1–20.

50. León-Cardenal E., Zúñiga-Galindo, W. A., "Local zeta functions for non-degenerate Laurent polynomials over p-adic fields," J. Math. Sci. Univ. Tokyo 20 (2013), no. 4, 569–595.

51. Chacón-Cortes L. F., Zúñiga-Galindo W. A., "Nonlocal Operators, Parabolic-type Equations, and Ultrametric Random Walks," J. Math. Phys. 54, 113503 (2013). 4. Erratum 55, 109901 (2014).

52. Casas-Sánchez O., Zúñiga-Galindo W. A., "Riesz Kernels and Pseudo-differential Operators Attached to Quadratic Forms Over p-adic Fields," p-Adic Numbers, Ultrametric Analysis, and Applications 5 (2013), no. 3, 177-193.

53. Torba S. M., Zúñiga-Galindo W. A., "Parabolic Type Equations and Markov Stochastic Processes on Adeles,"J. Fourier Anal. Appl. 19 (2013), no. 4, 792-835.

54. E. León-Cardenal, Veys, W., Zúñiga-Galindo W.A., "Poles of Archimedean Zeta Functions for Analytic Mappings," J. Lond. Math. Soc. (2) 87 (2013), no. 1, 1-21.

55. Galeano-Peñaloza J., Zúñiga-Galindo W. A., "Pseudo-differential operators with semi-quasielliptic symbols over p-adic fields," J. Math. Anal. Appl. 386 (2012), no. 1, 32–49.

56. Zúñiga-Galindo W.A., " Local zeta functions and fundamental solutions for pseudo-differential operators over p-adic Fields," p-Adic Numbers, Ultrametric Analysis, and Applications 3 (2011), no. 4, 344-358.

57. Segers D., Zúñiga-Galindo W.A., "Exponential sums Along p-adic Analytic Submanifolds," Finite Fields and its Applications 17 (2011), no. 4, 303-316.

58. Moyano-Fernandez J. J., Zúñiga-Galindo W.A., "Motivic Zeta Functions for Curve Singularities," Nagoya Math. J. 198 (2010), 47-75.

59. Rodriguez-Vega J. J., Zúñiga-Galindo W. A., "Elliptic Pseudo-Differential Equations and Sobolev Spaces over p-adic Fields," Pacific J. Math. 246 (2010), no. 2, 407–420.

60. Zúñiga-Galindo W.A., "Local Zeta Functions Supported on Analytic Sets and Newton Polyhedra," Int. Math. Res. Not. IMRN 2009, no. 15, 2855–2898.

61. Rodríguez-Vega, J. J.; Zúñiga-Galindo, W. A., "Taibleson operators, p-adic parabolic equations and ultrametric diffusion," Pacific J. Math. 237 (2008), no. 2, 327-347.

62. Zúñiga-Galindo W.A., "Parabolic Equations and Markov Processes Over p-adic Fields," Potential Anal. 28, (2008), 185-200.

63. Veys W., Zúñiga-Galindo W.A, "Zeta functions associated with polynomial mappings, log-principalization of ideals, and Newton polyhedra," Trans. Amer. Math. Soc. 360 (2008), no. 4, 2205–2227.

64. Zuniga-Galindo W.A., "Multiparametric exponential sums associated with semiquasi-homogeneous polynomial mappings," Finite Fields Appl. 13 (2007), no. 4, 936-945.

65. Zuniga-Galindo W.A., "Decay of Solutions of Wave-Type Pseudo-Differential Equations over p-adic Fields," Publications of RIMS, Vol 42 (2005), 461-479; Errata 44 (2008), no. 1, 45-48.

66. Zuniga-Galindo W.A., "Local zeta function for non-degenerate homogeneous mappings," Pacific J. Math. 218 (2005), no 1, 187-200.

67. Saia M.J., and Zuniga-Galindo W.A., "Local zeta functions for curves, non-degeneracy conditions and Newton polygons," Trans. Amer. Math. Soc. 357 (2005), 59-88.

68. Zuniga-Galindo W.A., "Pseudo-differential equations connected with p-adic forms and Igusa's local zeta functions," Bull. Aust. Math. Soc. 70 (2004), 73-86.

69. Zuniga-Galindo W.A., "On the poles of Igusa's local zeta function for algebraic sets," Bull. London Math. Soc. 36 (2004), 310-320.

70. Zuniga-Galindo W.A., "p-adic oscillatory integrals and Newton polyhedra, In Memoriam of Professor Jairo Charris," Rev. Acad. Colomb. Cienc. Exactas Fís. Nat. Vol 28 (106) (2004), 95-99.

71. Zuniga-Galindo W.A., "Computing Igusa's local zeta functions of univariate polynomials, and linear feedback shift registers," J. Integer Seq. Vol 6(2003), Article 03.3.6.

72. Zuniga-Galindo W.A., "Local zeta functions and Newton Polyhedra," Nagoya Math. J., Vol 172 (2003), 31-58.

73. Zuniga-Galindo W.A. "Fundamental solutions of pseudo-differential operators over p-adic fields," Rend. Sem. Mat. Univ. Padova 109 (2003), 241-245.

74. Zuniga-Galindo, W. A. "Exponential sums along p-adic curves," Finite Fields Appl. 9 (2003), no. 2, 140-149.

75. Zúñiga-Galindo W. A. "Igusa's local zeta functions of semiquasihomogeneous polynomials," Trans. Amer. Math. Soc. 353 (2001), no. 8, 3193-3207.

76. Zúñiga Galindo W. A., "Number of rational points of a singular curve," Proc. Amer. Math. Soc. 126 (1998), no. 9, 2549–2556.

77. Zúñiga-Galindo W. A., "Zeta functions of singular curves over finite fields," Rev. Colombiana Mat. 31 (1997), no. 2, 115-124.

78. Zúñiga Galindo W. A., "Zeta functions and Cartier divisors on singular curves over finite fields," Manuscripta Math. 94 (1997), no. 1, 75-88.

79. Zúñiga Wilson, "On dihedral algebraic function fields," Rev. Colombiana Mat. 27 (1993), no. 3-4, 213-222.

Papers Accepted

80. Zúñiga-Galindo W. A., Chacón-Cortés L.F. "Continuous-Time Quantum Markov Chains And Discretizations Of p-Adic Schrödinger Equations: Comparisons And Simulations." Accepted in the Journal of Statistical Mechanics: Theory and Experiment. arXiv:2508.06712.

Papers Under Revision

81. Zúñiga-Galindo W. A., "Critical Organization of Deep Neural Networks, and p-Adic Statistical Field Theories." arXiv:2601.19070. Under revision for the Journal of Physics A: Mathematical and Theoretical.

Papers Submitted

82. Zúñiga-Galindo W. A., "Hierarchical connectivity sets the upper critical dimension of neural avalanches." arXiv:2610.36404. Submitted to Physical Review Letters.

83. Zúñiga-Galindo W. A., "p-Adic statistical field theory of fluctuation effects in hierarchical neural networks." arXiv:2610.36400. Submitted to SciPost.

84. Zúñiga-Galindo W. A., "Wavefunctions localization, and the Wigner's Friend Paradox in a Framework of Discrete-Space Hypothesis." Submitted to Fortschritte der Physik. arXiv:2607.00198.

85. Zúñiga-Galindo W. A., "p-Adic Dirac Equations and the Jackiw-Rebbi Model." Submitted to Journal of Mathematical Physics. arXiv:2603.17200.

86. Zúñiga-Galindo W. A., "p-Adic Dirac Equations, Continuous-Time Quantum Walks, and Quantum Networks." arXiv:2609.04358. Submitted to Chaos, Solitons & Fractals.

Expository and Survey Articles

87. Albis Víctor, Zúñiga-Galindo Wilson, "An elementary introduction to the theory of Igusa local zeta functions," (Spanish) Lect. Mat. 20 (1999), no. 1, 5-33.

88. Leonardo F. Chacón Cortés y Wilson A. Zúñiga Galindo, "Análisis no Arquimediano y Sistemas Complejos," in universo.math.

89. León-Cardenal Edwin, Zúñiga-Galindo W. A., "An introduction to the theory of local zeta functions from scratch," Rev. Integr. Temas Mat. 37 (2019), no. 1, 45–76.

Selected Invited Presentations (2022–2026)

- (10/2026) p-Adic Quantum Mechanics, Non-locality, and the Measurement Problem. International Conference "New Horizons in Mathematical Physics," dedicated to the 80th Anniversary of Professor Igor Volovich.
- (6/2026) Local Zeta Functions, String Amplitudes, and Graphs. Sixth International Workshop on Zeta Functions in Algebra and Geometry. 8–12 June 2026, University of Galway, Ireland.
- (6/2026) Quantum Mechanics, Non-locality, and the Space Discreteness Hypothesis. IARD 2026. The 15th Biennial Conference on Classical and Quantum Relativistic Dynamics of Particles and Fields. 8 - 11 June 2026, Howard University, Washington DC.
- (2/2026) Continuous-Time Quantum Markov Chains and p-Adic Quantum Mechanics. Probability and Mathematical Physics Seminar at UC Riverside (UCR), February 10, 2026.
- (12/2025) Statistical Field Theory of Deep Neural Networks: An Ultrametric Approach to Deep Learning. Miami Physics Conference 2025, Dec 12-18, 2025.
- (11/2025) Continuous-Time Quantum Markov Chains and Discretizations of p-Adic Schrödinger Equations: Comparisons and Simulations. Tercera Escuela sobre la Física de los Materiales Cuánticos, November 4-6, 2025, Tuxtla Gutiérrez, Chiapas, México.
- (05/2025) p-Adic Quantum Mechanics, Quantum Walks, and the Discreteness of Space.NONLINEARITY, NONLOCALITY AND ULTRAMETRICITY. International Conference on the Occasion of Branko Dragovich 80th Birthday. 26 - 30/05/2025, Belgrade, Serbia.
- (03/2025) p-Adic numbers, neural networks, deep learning, and statistical field theories. Tercer Encuentro Nacional de Análisis no Arquimediano, Física Matemática y Aplicaciones, Tunja Colombia, March 17-20, 2025.

- (02/2025) p-Adic Boltzmann Machines. Montréal - Guanajuato Workshop on Probability and Machine Learning, Guanajuato, Mexico, February 25-27, 2025.

- (11/2024) The p-Adic Schrödinger Equation and the Two-Slit Experiment in Quantum Mechanics. Physics and Its Applications, PHYSICS 2024, Boston, October 21-24, 2024.

- (06/2024) Quantum Mechanics at the Planck Length. Sixteenth International Conference Application of Mathematics in Technical and Natural Sciences, AMiTaNS24. June 21-26, 2024, Albena, Bulgaria.

- (04/2024) A Correspondence Between Deep Boltzmann Machines and p-Adic Statistical Field Theories. Spring Eastern Sectional Meeting, Special Session on New Trends in Mathematical Physics, Howard University, Washington, DC, April 6-7, 2024.

- (04/2024) p-Adic Numbers, Neural Networks, Deep Learning, and Statistical Field Theories.Spring Eastern Sectional Meeting, Special Session on Artificial Intelligence Emergent From Mathematics and Physics, Howard University, Washington, DC, April 6-7, 2024.

- (03/2024) p-Adic Numbers, Neural Networks, deep learning, and Statistical Field Theory: An introduction. Coloquio de Matemática. Xavierian Pontifical University, Bogotá, Colombia.

- (04/2023) p-Adic Statistical Field Theory and Deep Boltzmann Machines. Workshop on Applied Analysis and Computation, School of Mathematical and Statistical Sciences, The University of Texas Rio Grande Valley (UTRGV).

- (11/2022) p-Adic Neural Networks and Quantum Field Theory. International Conference "New Trends in Mathematical Physics, " (November 7–12, 2022). Steklov Mathematical Institute of Russian Academy of Sciences, Moscow, Russia.

- (11/2022) p-Adic Statistical Field Theory and Deep Belief Networks.5th Annual Meeting, SIAM TX-LA Section, November 4–6, 2022, University of Houston, Houston, TX. USA.

- (10/2022). Graphs, local zeta functions, log-Coulomb gases, and phase transitions at finite temperature. LAGARTOS =Latin American Geometría Algebraica Real and Tropical Online Seminar.
- (07/2022) Ultrametric Diffusion, Rugged Energy Landscapes and Transition Networks. Physics and its Applications, July 18-21, San Francisco, CA, United States.
- (05/2022) Mini course "Introduction to p-adic analysis." CIMPA-CIMAT Research School: p-adic Numbers, Ultrametric Analysis, and Applications, Guanjuato, Mexico.
- (05/2022) Ultrametric diffusion, rugged energy landscapes and transition networks. Coloquio Virtual FMAT-CIMAT. Mexico.
- (03/2022) Ultrametricity: perspectives. International Seminar of Mathematics: Juntos UTRGV-UNAL. United States.

Ph.D. Dissertations Supervised

1. Chayapuntika Indoung, "Pattern Formation in Quantum and Classical Hierarchical Cellular Neural Networks," University of Texas Rio Grande Valley, 2026.
2. Baboucarr Dibba, "p-Adic Cellular Neural Networks With Delay," University of Texas Rio Grande Valley, 2025.
3. Nathaniel Mayes, "p-Adic Quantum Mechanics, Infinite Potential Wells, and Continuous-Time Quantum Walks," University of Texas Rio Grande Valley, 2025.
4. Carlos Alberto Garcia Bibiano, "p-Adic FitzHugh-Nagumo Systems and Turing Patterns," Cinvestav-I.P.N, Mexico, 2023.
5. Brian Andres Zambrano Luna, "p-Adic Cellular Neural Networks and Applications." Cinvestav-I.P.N, Mexico, 2023.
6. Angela Rocío Fuquen Tibatá, "p-Adic Open String Amplitudes and Quantum Field Theory." Cinvestav-I.P.N, Mexico, 2023.

7. Víctor Antonio Aguilar Arteaga, " Contributions to the Study of the Pseudodifferential Equations over Ultrametric Spaces." Cinvestav-I.P.N, Mexico, 2019.

8. María Luisa Martínez Mendoza, "Quantum field theory of scalar fields on a p-adic space-time." Cinvestav-I.P.N, Mexico, 2019. Other advisor Dr. José Antonio Vallejo.

9. Edilberto Arroyo Ortiz, "Construction of p-adic Quantum Fields in the Framework of White Noise Analysis." Cinvestav-I.P.N, Mexico, 2019.

10. Miriam Bocardo Gaspar, Local Zeta Functions and p-adic String Amplitudes." Cinvestav-I.P.N, Mexico, 2017.

11. Adriana Alexandra Albarracín Matilla, "Igusa's Local Zeta Functions and Exponential Sums for Arithmetically Non Degenerate Polynomials." Cinvestav-I.P.N, Mexico, 2017. Other advisor Dr. E. León Cardenal.

12. Anselmo Torresblanca Badillo, "Ultrametric Diffusion, Energy Landscapes, Feller Semigroups, and Pseudodifferential Operators Over p-adics." Cinvestav-I.P.N, Mexico, 2017.

13. Leonardo Fabio Chacón Cortés, "Non-local operators, Pseudodifferential Equations of Parabolic Type With Variable Coefficients and Markov Processes over p-adics." Cinvestav-I.P.N, Mexico, 2014.

14. Oscar Francisco Casas Sánchez, "Local Zeta Functions, Functional Equations and Pseudodifferential Operators Over p-adic Fields." National University of Colombia, Bogota, 2014.

15. Edwin León Cardenal, "Local Zeta Functions for Analytic Mappings, Non-degeneracy Conditions, and Newton Polyhedra." Cinvestav-I.P.N, Mexico, 2013.

16. Jeanneth Galeano Peñaloza, "Pseudo-differential Equations With Quasi-elliptic Symbols Over p-adic Fields ." Cinvestav-I.P.N, Mexico, 2012.

17. John J. Rodriguez Vega, "Local Zeta Functions and Pseudo-Differential Equations Over p-adic Fields," National University of Colombia, Bogota, 2010. Other advisor Dr. V. Albis González.

Postdoctoral Researchers Supervised

Dr. Samuel Estala Arias, Postdoctoral Fellow supported by Conacyt, Mexico (2014–2015).

Selected Organized Conferences

Ten of twenty-two international meetings, schools, and workshops organized or co-organized since 2007.

1. (Jointly with Tristan Hubsch) Special Session on New Trends in Mathematical Physics in 2024 Spring Eastern Sectional Meeting, Howard University, Washington, DC, April 6-7, 2024.

2. CIMPA-CIMAT Research School: p-Adic Numbers, Ultrametric Analysis, and Applications. Guanajuato, Mexico, Mayo 23-31, 2022 (jointly with Trond Digernes).

3. Fifth International Workshop on Zeta Functions In Algebra And Geometry, May 2-6, 2022, Nice, France (jointly with Alejandro Melle Hernandez, Adam Parusiński, Wim Veys, Ann Lemahieu).

4. (Conference's Chairman) Eighth International Conference on p-Adic Mathematical Physics and its Applications, Web Conference, May 17-28, 2021. YouTube Channel of the Conference.

5. First Encounter in Mathematical Biology and p-Adic Methods, (Jointly with Yuriria Cortés Poza, IIMAS, UNAM-Yucatán, and J. Rogelio Pérez Buendía, CIMAT-Mérida.) November 11-13, 2019, CIMAT, Merida, Mexico.

6. p-Adic Analysis, Arithmetic and Singularities, L. Santaló Summer Research School 2019 (Jointly with A. Melle Hernandez, C. Galindo Pastor, J. J. Moyano Fernández), June 24-29, 2019, Santander, Spain.

7. Mini-Workshop on p-Adic Mathematical Physics (Jointly with Sergii Torba), Querétaro, May 13th-15th, 2019.

8. (Conference's Chairman) Sixth International Conference on p-adic Mathematical Physics and Its Applications, Mexico.p-adics2017, Mexico City, October 23rd-27th, 2017.

9. (Jointly with Benjamin Klopsch, Willem Veys, Christopher Voll)Fourth International Workshop on Zeta Functions in Algebra and Geometry, Bielefeld, Germany, May 29th to June 2nd 2017.

10. Organizer of the Special Session of Non-Archimedean Analysis (jointly with Przemyslaw Górka, Alexander Quintero, Enrique Reyes García), in V Congreso Latinoamericano de Matemáticos, Barranquilla, Colombia, July 11th to 15th, 2016.

Editorial and Professional Service

Editorial boards. Scientific Reports (2025–); Axioms (2024–); Turkish Journal of Mathematics (2021–); Journal of Nonlinear Mathematical Physics (2021–); p-Adic Numbers, Ultrametric Analysis, and Applications (2016–); Revista Colombiana de Matemáticas (2011–).

Refereeing. Referee for Proceedings of the Royal Society A; Pacific Journal of Mathematics; Forum Mathematicum; Annals of Physics; Journal of Mathematical Physics; Nuclear Physics B; Journal of Mathematical Analysis and Applications; Machine Learning: Science and Technology; Neurocomputing; Scientific Reports; Physica A; Physica D; Chaos, Solitons and Fractals; among others.

External doctoral committees. Karlsruhe Institute of Technology, Germany (2026); University of Camerino, Italy (2024).

Research seminar. Founder and organizer of the bi-weekly Virtual Research Seminar on Non-Archimedean Analysis and Mathematical Physics, which has run since 2021.

Teaching and Curriculum Development

More than twenty-seven years of university teaching in Colombia, Mexico, and the United States; over thirty distinct courses. At UTRGV (2021–2026) I have taught fifteen distinct courses across all levels, from Calculus and Linear Algebra through Topology and Complex Variables to graduate Analysis, Algebra, and Analytic Number Theory. I designed and taught **MATH 8335, Deep Learning** (doctoral, 2024), built around the mathematical foundations of deep networks rather than programming frameworks, with individual research projects in place of examinations. At CINVESTAV (2007–2020)

I taught graduate courses in Functional Analysis, Real Analysis, Algebraic Geometry, p-Adic Analysis, and Topics in Local Zeta Functions.